7.1.2 Cell Shape and Arrangement of Bacteria

The shape of a bacterium is governed by its rigid cell wall. Typical bacterial cells are spherical (called cocci), straight rods (called bacilli) and helically curved rods (called spiral). These shapes are constant for the particular species or genus but there are bacterial cells that are pleomorphic in nature. They exhibit a variety of shapes.

- Cocci appear in several characteristic arrangements, depending on the plane of cellular division and whether daughter cells remain together with the parents even after cell division. The cells may occurs in pairs (diplococci), in groups of four (tetracocci), in clusters (*Staphylococcus*), in a bead like chain (*Streptococci*) or in cuboidal arrangement of cells (*Sarcinae*).
- Bacilli are rod shaped organism (Singular, bacillus = stick) usually ranging between 1 and 10 µm in length. Some bacilli are so short and stumpy that they appear ovoid and are referred to as coccobacilli. Bacilli are not arranged in patterns as complex as those of cocci and mostly occur as singles or in pairs (diplobacilli, Example: *Bacillus subtilis)* or in the form of chains (*Streptobacilli*). Some form trichomes, which are similar to chains. In other *Bacilli* such as *Corynebacterium diphtheria* the cells are lined side by side like matchsticks (pallisade arrangement). Some bacilli are curved into a form resembling a comma. These cells are called vibrios as in *Vibrio cholera*.
- Spiral bacteria: They are divided into two groups, spirilla (singular spirillum) and spirochetes (agent of syphilis). Although these two are similar in shape spirochetes are flexible in nature. Spiral bacteria are far too thin to be seen with the standard Brightfield microscope but are readily observed by Darkfield microscope (Figure 7.3).

Filamentous bacteria

Bacteria tend to form long strands composed of many cells. In these cases, an occasional single cell may be seen after it breaks away from a long filament. These organisms resemble the threadlike strands of fungi but their internal structure is typical of bacteria. Filamentous soil bacteria include *Streptomyces* species.

Pleomorphic bacteria

A few bacteria lack rigid cell walls, and their flexible plasma membrane allows them to change shape. These are called pleomorphic bacteria (pleo-more; morph-form). Example: *Mycoplasma*.

7.2 Structures External to Cell Wall of Bacetria

7.2.1 Appendages

Flagella

Flagella (singular flagellum) are threadlike, long, thin helical filaments measuring 0.01-0.02nm in diameter. These appendages extend outward from the plasma membrane and cell wall. Flagella are so thin that they cannot be observed directly with a bright field microscope, but must be stained with special techniques (example: Fontana's silver staining technique) that increase their thickness. The detailed structure of a flagellum can only be seen in the electron microscope.

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